Disaster Management: Relief Distribution

Using Graph CRUD C

Data Structures — Final Project Report

Team Members

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Abstract:

Disaster management is an essential process aimed at reducing the impact of natural calamities such as floods, earthquakes, and cyclones. One of the most critical challenges during such events is the efficient distribution of relief resources to affected areas. This project presents a Relief Distribution Management System using a Graph Data Structure, where locations are represented as vertices, routes are represented as edges, and relief quantities are assigned as weights. The system utilizes an adjacency matrix representation to efficiently manage the connections between different locations. Various operations such as adding, deleting, updating, searching, and displaying routes are implemented. This approach ensures faster decision-making and improved coordination in disaster scenarios.

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1. Introduction:

Natural disasters often disrupt infrastructure, making it challenging to deliver essential supplies. Efficient planning and management are required to ensure that relief materials reach affected areas promptly. Graph theory provides an effective way to model such scenarios. In this project, a graph-based approach is used where nodes represent locations and edges represent connections with associated relief quantities. The adjacency matrix is chosen for its simplicity and ease of implementation. This system demonstrates how data structures can be applied to solve real-world problems in disaster management.

2.Objectives:

The main objective of this project is to design and implement a system that manages relief distribution efficiently. It aims to represent locations and routes using a graph, allow dynamic updates to the network, and provide quick access to route information. The system also focuses on practicing fundamental programming concepts such as dynamic memory allocation and structured programming in C.

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3. Scope of the Project

This project is limited to small-scale simulation of relief distribution networks. It can be used in academic environments to understand graph implementation and operations. Although it does not include real-time data integration or GPS tracking, it lays the foundation for scalable disaster management systems. Future enhancements can integrate

real-world mapping and optimization algorithms.

4. Literature Review

Graph data structures have been widely used in transportation, network routing, and disaster management systems. Previous studies show that adjacency matrices are effective for dense graphs where connections between nodes are frequent. While adjacency lists are more memory efficient, adjacency matrices provide faster access to edge information. Similar systems in disaster response use graph traversal and shortest path algorithms to optimize relief delivery. This project builds upon these concepts with a simplified implementation.

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5. Methodology

The system is developed using a structured approach. First, the graph is initialized with a specified number of vertices. Memory is dynamically allocated for the adjacency matrix. Users interact with the system through a menu-driven interface. Based on user input, operations such as adding, deleting, updating, and searching routes are performed. Each operation directly modifies or queries the adjacency matrix.

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6. Technology Used

The system is implemented using the C programming language. Standard libraries such as stdio.h and stdlib.h are used for input/output operations and memory management. The adjacency matrix is implemented using double pointers to dynamically allocate memory. The program runs in a console-based environment.

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7. Architecture / System Design

The system follows a modular design approach. It consists of separate functions for each operation such as graph creation, edge manipulation, and display. The adjacency matrix serves as the core data structure. The user interface is menu-driven, ensuring ease of interaction. Data flows from user input to processing functions and finally to output display.

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8. Implementation Details

The implementation begins with defining a Graph structure containing the number of vertices and a pointer to the adjacency matrix. The createGraph function dynamically allocates memory and initializes the matrix with zeros. The addEdge function creates a connection between two nodes with a specified weight. The updateEdge function modifies the weight of an existing edge, while deleteEdge removes a connection. The searchEdge function checks whether a route exists and displays its weight. The displayGraph function prints the adjacency matrix. A menu-driven main function allows users to interact with the system continuously until they choose to exit.

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9. Results / Output

The system successfully demonstrates all graph operations. Users can add routes

between locations, update relief quantities, delete routes, and search for specific connections. The adjacency matrix display provides a clear visualization of the network. The program handles user inputs effectively and updates the graph dynamically. The output confirms correct implementation of graph operations.

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10. Conclusion

This project demonstrates the practical application of graph data structures in disaster management. The adjacency matrix representation simplifies implementation and allows efficient operations on the graph. The system provides a basic yet effective simulation of relief distribution. It enhances understanding of data structures and their real-world applications.

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11. Future Work

The project can be enhanced by implementing advanced algorithms such as Dijkstra's algorithm for shortest path optimization. Integration with real-time data sources and GPS systems can make it more practical. A graphical user interface can improve usability. Additionally, converting the system into a web-based or mobile application would increase accessibility and scalability.

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12. Footnotes / References

Thomas H. Cormen et al., Introduction to Algorithms, MIT Press.

Ellis Horowitz, Fundamentals of Data Structures in C.

Data Structures and Algorithms using C – Reema Thareja.

Research papers on disaster management systems using graph theory.

Online resources and documentation for C programming and graph implementation.